

MongoDB Complete Commands - Student Collection

1. Database Commands

```
[test> show dbs
admin          40.00 KiB
bookflow_db    132.00 KiB
config         36.00 KiB
local          40.00 KiB
[test> use studentDB
switched to db studentDB
[studentDB> db
studentDB
studentDB> █
```

2. Collection Commands

```
studentDB> show collections

studentDB> db.createCollection("students")
{ ok: 1 }
studentDB> show collections
students
studentDB> █
```

3. Sample Student Data

```
studentDB> db.students.insertMany([
|   { rollNo: 101, name: "John", age: 20, department: "CSE", marks: 85, city: "Hyderabad" },
|   { rollNo: 102, name: "Alice", age: 21, department: "CSE", marks: 90, city: "Vijayawada" },
|   { rollNo: 103, name: "Rahul", age: 20, department: "ECE", marks: 75, city: "Hyderabad" },
|   { rollNo: 104, name: "Priya", age: 22, department: "IT", marks: 88, city: "Warangal" },
|   { rollNo: 105, name: "Kiran", age: 21, department: "CSE", marks: 95, city: "Hyderabad" },
|   { rollNo: 106, name: "Anu", age: 20, department: "IT", marks: 82, city: "Guntur" },
|   { rollNo: 107, name: "Ravi", age: 23, department: "ECE", marks: 68, city: "Nellore" },
|   { rollNo: 108, name: "Sita", age: 21, department: "CSE", marks: 91, city: "Hyderabad" },
|   { rollNo: 109, name: "Arun", age: 22, department: "AI&DS", marks: 87, city: "Kurnool" },
|   { rollNo: 110, name: "Meena", age: 20, department: "AI&DS", marks: 93, city: "Hyderabad" }
| ])
| {
|   acknowledged: true,
|   insertedIds: {
|     '0': ObjectId('6a851e447cdcd38d9b0a284e'),
|     '1': ObjectId('6a851e447cdcd38d9b0a284f'),
|     '2': ObjectId('6a851e447cdcd38d9b0a2850'),
|     '3': ObjectId('6a851e447cdcd38d9b0a2851'),
|     '4': ObjectId('6a851e447cdcd38d9b0a2852'),
|     '5': ObjectId('6a851e447cdcd38d9b0a2853'),
|     '6': ObjectId('6a851e447cdcd38d9b0a2854'),
|     '7': ObjectId('6a851e447cdcd38d9b0a2855'),
|     '8': ObjectId('6a851e447cdcd38d9b0a2856'),
|     '9': ObjectId('6a851e447cdcd38d9b0a2857')
|   }
| }
studentDB> db.students.countDocuments()
10
```

4. CREATE - Insert Commands

```
studentDB> db.students.insertOne({
|   rollNo: 111,
|   name: "Vijay",
|   age: 21,
|   department: "CSE",
|   marks: 86,
|   city: "Hyderabad"
| })
{
  acknowledged: true,
  insertedId: ObjectId('6a851f077cdcd38d9b0a2858')
}

studentDB> db.students.insertMany([
|   { rollNo: 112, name: "Pooja", age: 20, department: "IT", marks: 84 },
|   { rollNo: 113, name: "Ramesh", age: 22, department: "ECE", marks: 79 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a851f9a7cdcd38d9b0a2859'),
    '1': ObjectId('6a851f9a7cdcd38d9b0a285a')
  }
}
studentDB> █
```

5. READ - Basic Find Commands

db.students.find()

```
studentDB> db.students.find()
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2850'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  }
]
```

db.students.findOne()

```
[studentDB> db.students.findOne()
{
  _id: ObjectId('6a851e447cdcd38d9b0a284e'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
```

db.students.findOne({ rollNo: 101 })

```
[studentDB> db.students.findOne({ rollNo: 101 })
{
  _id: ObjectId('6a851e447cdcd38d9b0a284e'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
studentDB>
```

db.students.find({ department: "CSE" })

```
[studentDB> db.students.find({ department: "CSE" })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  }
]
```

```
db.students.find({ name: "John" })
```

```
[studentDB> db.students.find({ name: "John" })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  }
]
```

```
db.students.countDocuments()
```

```
[studentDB> db.students.countDocuments()
13
studentDB> 
```

```
db.students.countDocuments({ department: "CSE" })
```

```
[studentDB> db.students.countDocuments({ department: "CSE" })
5
studentDB> 
```

6. Comparison Operators

// Greater than

```
[studentDB> db.students.find({ marks: { $gt: 80 } })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  }
]
```

// Greater than or equal

```
[studentDB> db.students.find({ marks: { $gte: 80 } })]
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  }
]
```

// Less than

```
[studentDB> db.students.find({ marks: { $lt: 80 } })]
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2850'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  }
]
```


// Less than or equal

```
[studentDB> db.students.find({ marks: { $lte: 80 } })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2850'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  }
]
```

// Not equal

```
[studentDB> db.students.find({ department: { $ne: "CSE" } })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2850'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  }
]
```

// Equal

```
studentDB> db.students.find({ marks: { $eq: 90 } })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  }
]
```

7. Logical Operators

// AND Operator

```
studentDB> db.students.find({
  $and: [
    { department: "CSE" },
    { marks: { $gt: 85 } }
  ]
})
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  }
]
```

// OR Operator

```
studentDB> db.students.find({
|   $or: [
|     { department: "CSE" },
|     { department: "IT" }
|   ]
| })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  }
]
```

// NOT Operator

```
studentDB> db.students.find({
|   marks: { $not: { $gt: 80 } }
| })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2850'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  }
]
```


// NOR Operator

```
studentDB> db.students.find({
  $nor: [
    { department: "CSE" },
    { marks: { $gt: 90 } }
  ]
})
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2850'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  },
]
```

8. IN and NIN Operators

// IN Operator

```
studentDB> db.students.find({
  department: { $in: ["CSE", "IT"] }
})
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  }
]
```

studentDB>

//NIN Operator

```
studentDB> db.students.find({
|   department: { $nin: ["CSE", "IT"] }
| })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2850'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  }
]
```

9. UPDATE Commands

// Update one

```
studentDB> db.students.updateOne(
|   { rollNo: 101 },
|   { $set: { marks: 92 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

// Update multiple

```
studentDB> db.students.updateMany(
|   { department: "CSE" },
|   { $set: { status: "Active" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

// Increment

```
studentDB> db.students.updateOne(
|   { rollNo: 101 },
|   { $inc: { marks: 5 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.find({ rollNo: 101 })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 97,
    city: 'Hyderabad',
    status: 'Active'
  }
]
```

// Decrement

```
studentDB> db.students.updateOne(
|   { rollNo: 101 },
|   { $inc: { marks: -2 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.find({ rollNo: 101 })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad',
    status: 'Active'
  }
]
```

// Rename a field

```
studentDB> db.students.updateMany(
|   {},
|   { $rename: { "city": "location" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 13,
  modifiedCount: 11,
  upsertedCount: 0
}
studentDB> db.students.find({ rollNo: 101 })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  }
]
```

// Add field

```
studentDB> db.students.updateOne(
|   { rollNo: 101 },
|   { $set: { attendance: 90 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.find({ rollNo: 101 })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad',
    attendance: 90
  }
]
```

// Remove a field

```
studentDB> db.students.updateOne(
|   { rollNo: 101 },
|   { $unset: { attendance: "" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.find({ rollNo: 101 })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  }
]
```

10. DELETE Commands

// Delete one

```
[studentDB> db.students.deleteOne({ rollNo: 103 })
{ acknowledged: true, deletedCount: 1 }
[studentDB> db.students.find({ rollNo: 103 })
```

// Delete multiple

```
[studentDB> db.students.deleteMany({ marks: { $lt: 40 } })
{ acknowledged: true, deletedCount: 0 }
studentDB> █
```

11. Sorting

// Ascending

```
[studentDB> db.students.find().sort({ marks: 1 })
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    location: 'Nellore'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    location: 'Guntur'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    location: 'Kurnool'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    location: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    status: 'Active',
    location: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  }
]
```


// Descending

```
[studentDB> db.students.find().sort({ marks: -1 })]
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    status: 'Active',
    location: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    location: 'Warangal'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    location: 'Kurnool'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    location: 'Guntur'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    location: 'Nellore'
  }
]
```

// Sort by department and marks

```
[studentDB> db.students.find().sort({ department: 1, marks: -1 })]
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    location: 'Kurnool'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    status: 'Active',
    location: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    location: 'Warangal'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    location: 'Nellore'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    location: 'Warangal'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    location: 'Guntur'
  }
]
```

studentDB> █

12. Limit and Skip

// First 5 students

```
studentDB> db.students.find().limit(5)
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284f'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    status: 'Active',
    location: 'Vijayawada'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    location: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    location: 'Guntur'
  }
]
```

// Top 3 students

```
studentDB> db.students.find().sort({ marks: -1 }).limit(3)
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a284e'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    location: 'Hyderabad'
  }
]
```

// Skip first 5

```
studentDB> db.students.find().skip(5)
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2854'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    location: 'Nellore'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2855'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 92,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2856'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    location: 'Kurnool'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2857'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f077cdcd38d9b0a2858'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a2859'),
    rollNo: 112,
    name: 'Pooja',
    age: 20,
    department: 'IT',
    marks: 84
  },
  {
    _id: ObjectId('6a851f9a7cdcd38d9b0a285a'),
    rollNo: 113,

```

// Skip 2 and show next 3

```
studentDB> db.students.find().skip(2).limit(3)
[
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2851'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    location: 'Warangal'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2852'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  },
  {
    _id: ObjectId('6a851e447cdcd38d9b0a2853'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    location: 'Guntur'
  }
]
```

13. Projection

// Only name and marks

```
studentDB> db.students.find(
|   {},
|   { _id: 0, name: 1, marks: 1 }
| )
[
  { name: 'John', marks: 95 },
  { name: 'Alice', marks: 90 },
  { name: 'Priya', marks: 88 },
  { name: 'Kiran', marks: 95 },
  { name: 'Anu', marks: 82 },
  { name: 'Ravi', marks: 68 },
  { name: 'Sita', marks: 91 },
  { name: 'Arun', marks: 87 },
  { name: 'Meena', marks: 93 },
  { name: 'Vijay', marks: 86 },
  { name: 'Pooja', marks: 84 },
  { name: 'Ramesh', marks: 79 }
]
```

// Name, department and marks

```
studentDB> db.students.find(
|   {},
|   { _id: 0, name: 1, department: 1, marks: 1 }
| )
[
  { name: 'John', department: 'CSE', marks: 95 },
  { name: 'Alice', department: 'CSE', marks: 90 },
  { name: 'Priya', department: 'IT', marks: 88 },
  { name: 'Kiran', department: 'CSE', marks: 95 },
  { name: 'Anu', department: 'IT', marks: 82 },
  { name: 'Ravi', department: 'ECE', marks: 68 },
  { name: 'Sita', department: 'CSE', marks: 91 },
  { name: 'Arun', department: 'AI&DS', marks: 87 },
  { name: 'Meena', department: 'AI&DS', marks: 93 },
  { name: 'Vijay', department: 'CSE', marks: 86 },
  { name: 'Pooja', department: 'IT', marks: 84 },
  { name: 'Ramesh', department: 'ECE', marks: 79 }
]
```

// Hide age

```
studentDB> db.students.find(
|   { },
|   { age: 0 }
| )
[
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2b2'),
    name: 'John',
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2b3'),
    name: 'Alice',
    department: 'CSE',
    marks: 90,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2b4'),
    name: 'Priya',
    department: 'IT',
    marks: 88,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2b5'),
    name: 'Kiran',
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2b6'),
    name: 'Anu',
    department: 'IT',
    marks: 82,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2b7'),
    name: 'Ravi',
    department: 'ECE',
    marks: 68,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2b8'),
    name: 'Sita',
    department: 'CSE',
    marks: 91,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2b9'),
    name: 'Arun',
    department: 'AI&DS',
    marks: 87,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2ba'),
    name: 'Meena',
    department: 'AI&DS',
    marks: 93,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2bb'),
    name: 'Vijay',
    department: 'CSE',
    marks: 86,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2bc'),
    name: 'Pooja',
    department: 'IT',
    marks: 84,
    status: 'Active',
    location: 'Bangalore'
  },
  {
    _id: ObjectId('5a55a7c0b2b2b2b2b2b2b2bd'),
    name: 'Ramesh',
    department: 'ECE',
    marks: 79,
    status: 'Active',
    location: 'Bangalore'
  }
]
```

14. Distinct Values

// All departments

```
[studentDB> db.students.distinct("department")
[ 'AI&DS', 'CSE', 'ECE', 'IT' ]
studentDB> █
```

// All cities

```
[studentDB> db.students.distinct("city")
[]
studentDB> █
```

// Departments with marks 80 or above

```
[studentDB> db.students.distinct("department")
[ 'AI&DS', 'CSE', 'ECE', 'IT' ]
[studentDB> db.students.distinct("city")
[]
studentDB> db.students.distinct(
|   "department",
|   { marks: { $gte: 80 } }
| )
[ 'AI&DS', 'CSE', 'IT' ]
studentDB> █
```

15. Index Commands

// Create index

```
[studentDB> db.students.createIndex({ rollNo: 1 })
rollNo_1
studentDB> █
```

// Create name index

```
[studentDB> db.students.createIndex({ name: 1 })
name_1
studentDB> █
```

// Compound index

```
studentDB> db.students.createIndex({ department: 1, marks: -1 })
department_1_marks_-1
studentDB> █
```

// Show indexes

```
[studentDB> db.students.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { rollNo: 1 }, name: 'rollNo_1' },
  { v: 2, key: { name: 1 }, name: 'name_1' },
  {
    v: 2,
    key: { department: 1, marks: -1 },
    name: 'department_1_marks_-1'
  }
]
studentDB>
```

// Drop name index

```
[studentDB> db.students.dropIndex("name_1")
{ nIndexesWas: 4, ok: 1 }
studentDB>
```

// Drop all indexes except _id

```
[studentDB> db.students.dropIndexes()
{
  nIndexesWas: 3,
  msg: 'non-_id indexes dropped for collection',
  ok: 1
}
studentDB>
```

16. Aggregation - Basic Commands

// Display all documents through aggregation

```
studentDB> db.students.aggregate([
|   {
|     $group: {
|       _id: "$department",
|       totalStudents: { $sum: 1 }
|     }
|   }
| ])
[
|   { _id: 'AI&DS', totalStudents: 2 },
|   { _id: 'ECE', totalStudents: 2 },
|   { _id: 'CSE', totalStudents: 5 },
|   { _id: 'IT', totalStudents: 3 }
| ]
studentDB>
```

// Students with marks > 80

```
studentDB> db.students.aggregate([
|   { $match: { marks: { $gt: 80 } } }
| ])
|
| [
|   {
|     _id: ObjectId('6a851e447cdcd38d9b0a284e'),
|     rollNo: 101,
|     name: 'John',
|     age: 20,
|     department: 'CSE',
|     marks: 95,
|     status: 'Active',
|     location: 'Hyderabad'
|   },
|   {
|     _id: ObjectId('6a851e447cdcd38d9b0a284f'),
|     rollNo: 102,
|     name: 'Alice',
|     age: 21,
|     department: 'CSE',
|     marks: 90,
|     status: 'Active',
|     location: 'Vijayawada'
|   },
|   {
|     _id: ObjectId('6a851e447cdcd38d9b0a2851'),
|     rollNo: 104,
|     name: 'Priya',
|     age: 22,
|     department: 'IT',
|     marks: 88,
|     location: 'Warangal'
|   },
|   {
|     _id: ObjectId('6a851e447cdcd38d9b0a2852'),
|     rollNo: 105,
|     name: 'Kiran',
|     age: 21,
|     department: 'CSE',
|     marks: 95,
|     status: 'Active',
|     location: 'Hyderabad'
|   },
|   {
|     _id: ObjectId('6a851e447cdcd38d9b0a2853'),
|     rollNo: 106,
|     status: 'Inactive'
|   }
| ]
```

// Department-wise student count

```
studentDB> db.students.aggregate([
|   {
|     $group: {
|       _id: "$department",
|       totalStudents: { $sum: 1 }
|     }
|   }
| ])
|
| [
|   { _id: 'AI&DS', totalStudents: 2 },
|   { _id: 'IT', totalStudents: 3 },
|   { _id: 'CSE', totalStudents: 5 },
|   { _id: 'ECE', totalStudents: 2 }
| ]
studentDB> █
```

// Department-wise average marks

```
studentDB> db.students.aggregate([
|   {
|     $group: {
|       _id: "$department",
|       averageMarks: { $avg: "$marks" }
|     }
|   }
| ])
|
| [
|   { _id: 'ECE', averageMarks: 73.5 },
|   { _id: 'CSE', averageMarks: 91.4 },
|   { _id: 'IT', averageMarks: 84.66666666666667 },
|   { _id: 'AI&DS', averageMarks: 90 }
| ]
```

// Highest marks

```
studentDB> db.students.aggregate([
|   {
|     $group: {
|       _id: null,
|       highestMarks: { $max: "$marks" }
|     }
|   }
| ])
| [ { _id: null, highestMarks: 95 } ]
studentDB> █
```


// Lowest marks

```
studentDB> db.students.aggregate([
|   {
|     $group: {
|       _id: null,
|       lowestMarks: { $min: "$marks" }
|     }
|   }
| ])
[ { _id: null, lowestMarks: 68 } ]
studentDB>
```

// Total marks

```
studentDB> db.students.aggregate([
|   {
|     $group: {
|       _id: null,
|       totalMarks: { $sum: "$marks" }
|     }
|   }
| ])
[ { _id: null, totalMarks: 1038 } ]
studentDB>
```

17. Aggregation with Multiple Stages

// CSE students, grouped and sorted by average marks

```
studentDB> db.students.aggregate([
|   { $match: { department: "CSE" } },
|   {
|     $group: {
|       _id: "$department",
|       averageMarks: { $avg: "$marks" },
|       totalStudents: { $sum: 1 }
|     }
|   },
|   { $sort: { averageMarks: -1 } }
| ])
[ { _id: 'CSE', averageMarks: 91.4, totalStudents: 5 } ]
studentDB>
```

// Top 3 students using aggregation

```
studentDB> db.students.aggregate([
|   { $sort: { marks: -1 } },
|   { $limit: 3 },
|   {
|     $project: {
|       _id: 0,
|       rollNo: 1,
|       name: 1,
|       marks: 1
|     }
|   }
| ])
[
|   { rollNo: 105, name: 'Kiran', marks: 95 },
|   { rollNo: 101, name: 'John', marks: 95 },
|   { rollNo: 110, name: 'Meena', marks: 93 }
| ]
studentDB>
```